

Acidity-Mediated Metal Oxide Heterointerfaces: Roles of Substrates and Surface Modification

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Abstract

Although strong modulation of interfacial electron concentrations by the relative acidity of surface additives has been suggested, direct observation of corresponding changes in surface conductivity, crucial for understanding the role of local space charge, has been lacking. Here, we introduce a model platform comprising well-aligned mixed ionic-electronic conducting $\text{Pr}_{0.2}\text{Ce}_{0.8}\text{O}_{2-\delta}$ nanowire arrays (PCO_{NA}) to show that acidity-modulated heterointerfaces predict electron depletion or accumulation, resulting in tunable electrical properties. We confirm three orders of magnitude increased PCO_{NA} conductivity with basic Li_2O infiltration. Moreover, the relative acidity of the insulating substrate supporting the PCO_{NA} strongly influences its electronic properties as well. This strategy is further validated in purely ionic-conducting nanostructured ceria as well as PCO_{NA} . We suggest that observed conductivity changes stem not only from acidity-mediated space charge potentials at heterointerfaces but also from grain boundaries, chemically-modulated by cation in-diffusion. These findings have broad implications for how substrate and surface treatment choices can alter the conductive properties of nanostructured functional oxides.

Introduction

Heterointerfaces have long been recognized as having a profound influence on the electronic properties of semiconducting materials and their prudent engineering has been vital for the development of a broad variety of microelectronic devices including diodes, transistors, and solar cells¹⁻³. This stems from the ability of heterointerfaces to induce band bending, modify charge carrier distributions, and create built-in electric fields that govern the electronic behavior at interfaces^{4,5}. The same can be said for ionic materials, where heterointerfaces can lead to extraordinary properties arising from space charge effects with resultant defect redistribution. For example, the formation of two-dimensional electron gases at the heterointerfaces of insulating oxides and the emergence of superconductivity of relevance for quantum applications⁶⁻⁹ have been, in part, associated with the accumulation of oxygen vacancies acting as electron donors. In more conventional systems, these characteristics can be highly beneficial for enhancing the performance of energy storage and chemical conversion applications, including electrocatalysis, fuel cells, and batteries^{10,11}.

One representative work by Sata et al. demonstrated marked increases in ionic conductivity with increases in interface density of CaF₂ and BaF₂ heterostructures¹². By reducing the heterointerface spacing to below 50 nm, they observed the predicted mesoscopic size effect and revealed that the overlap of space charge regions at nanoscale interfaces can lead to anomalous transport behavior. These findings highlight that precise engineering of the heterointerface enables modulation of the flow of ionic as well as electronic charges. However, studies that successfully exemplify the role of heterointerface space charge in controlling electronic or ionic transport in metal oxides remain limited, except for a few restricted specific compositions¹³. This is due to difficulties in identifying material combinations with appropriate space charge characteristics and the inherent challenge of distinguishing its role from coexisting factors, such as strain^{14,15}, atomic reconstruction¹⁶, or adsorbate-induced charge localization¹⁷⁻¹⁹ at heterointerfaces. Recently, Steinbach et al. developed a physicochemical model to predict space charge properties at heterointerfaces between various oxides and undoped SrTiO₃ single crystals²⁰. They were able to relate the experimentally measured space charge potential to defect thermodynamics and the reducibility of oxides. While offering a widely applicable framework, the model depends highly on detailed knowledge of defect formation energies and reducibility of oxides, not always readily available. Moreover, their analysis employs mixed ionic and electronic conducting (MIEC) materials requiring gas-phase